

Leonardo De Andrade Santos

Beaconsfield, Sydney NSW 2015, Australia | leo.santos.engineering@gmail.com | +61 415 846 213

linkedin.com/in/leonardo-20-as | github.com/Je-Leo-AS

5 October 2026

Hiring Team
Silicon Quantum Computing
Sydney, NSW

Dear Hiring Team,

I am writing to apply for the Software Engineer, Control & Error Correction role at Silicon Quantum Computing. The role is especially interesting to me because it sits at the boundary between software, digital logic and physics-constrained realtime systems: exactly the kind of engineering environment where correctness, latency and measured behaviour matter more than abstract software architecture alone.

I am an embedded software and FPGA-oriented engineer with 4+ years of experience building production C/C++ and Python systems close to hardware, together with master's research in realtime DSP and VHDL implementation. My current research focuses on Digital Predistortion for RF power-amplifier linearisation, where I develop Python reference models, translate mathematically defined algorithms into fixed-point VHDL datapaths, and validate hardware-oriented implementations against bit-accurate simulation results. This has given me practical experience with deterministic processing, complex arithmetic, delay-line memory, pipelining, coefficient quantisation and the trade-offs between latency, precision and hardware resources.

In industry, I have worked as an Embedded & Software Engineer at IoTag Technology, developing low-level C/C++ software, Linux-based tooling and Python validation workflows for telemetry and equipment-monitoring systems. That work required debugging hardware/software interactions, communication failures and performance issues across deployed embedded systems, and translating real device behaviour into reliable software pipelines. I am comfortable working across firmware, Linux tooling and backend integration, and I value direct measurement on target systems rather than relying only on assumptions.

The SQC role's emphasis on realtime scheduling, hardware-facing interfaces, simulation, reference models, recorded device data and measured latency aligns strongly with how I approach engineering. While I have not yet worked directly on quantum error correction in production, I have a strong mathematical and signal-processing background, experience implementing algorithms from models into hardware-aware systems, and the discipline to reason about worst-case behaviour rather than average-case performance. I would be excited to bring that foundation to the realtime quantum control plane and to learn from physicists, quantum information scientists and platform engineers building the system.

I am based in Sydney and would welcome the opportunity to discuss how my embedded software, FPGA/VHDL and realtime DSP background could contribute to SQC's Control & Error Correction team.

Sincerely,

Leonardo De Andrade Santos